MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE:CLASS A ACCOMMODATION
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
Typ.	Typical
HCS	Hollow Core Slab
Fc'	Concrete Compressive Strength
Fy	Yield Strength of Reinforcing Steel
ϕ	Strength Reduction Factor
SBC	Saudi Building Code
SASO	Saudi Standards, Metrology and Quality Organization
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
PCI	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of an Accommodation Building (Class-A).

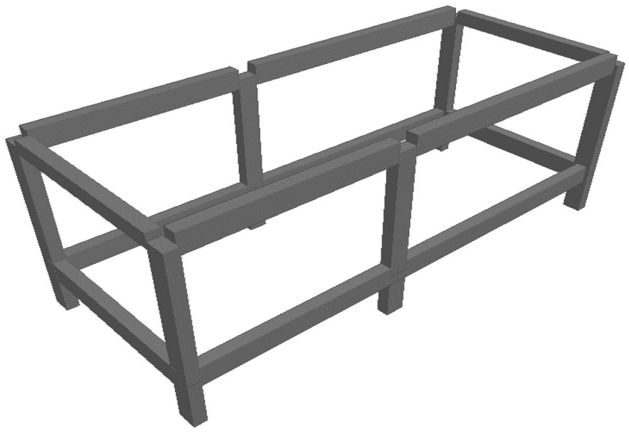
2. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Building-A only.

3. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.



STAAD 3D MODEL

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3.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

3.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

3.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

4. DESIGN LOADS

4.1. Dead Load

4.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-150 Topped) = **2.52 KN/m2**

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Prestressed Hollowcore Floor

Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load = **5.82 kN/m²**

UDL on Main Beams (LB1) = 5.82 x 4.2/2 =**12.22 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

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4.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 2. 5m clear height = **10.0 kN/m**

4.2. Live Load

4.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat = **1.00 kN/m²**

Udl Due to Live Load on the Ledger Beam (LB1) = 1.0X4.2/2 = **2.1 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 1*0.6*0.6/2 = **0.18 kN/m /m**

4.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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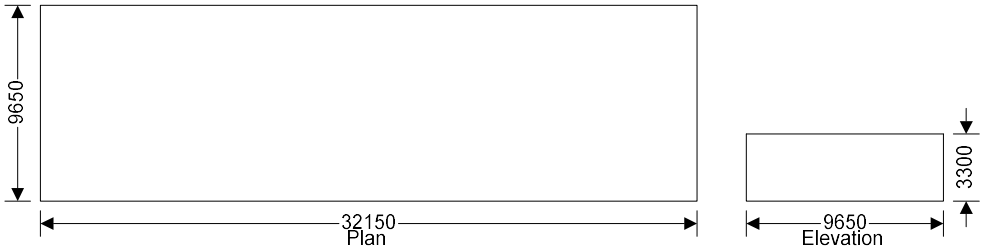
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4.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 10300 mm
Width of building;	d = 4200 mm
Height to eaves;	H = 3300 mm
Mean height;	h = 3300 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = 0.85(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

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Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.); q_i = 1.289 kN/m^2

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	34.00	5.70
B	3.30	-0.50	1.29	-1.26	33.99	-42.70
C	3.30	-0.70	1.29	-1.48	13.86	-20.45
D	3.30	-0.70	1.29	-1.48	13.86	-20.45

Overall loading

Projected vertical plan area of wall;

$A_{vert_w_0} = b \times H = 33.99$ m^2

Projected vertical area of roof; $A_{vert_r_0}$ = 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0}$ = 26.2 kN

Leeward net force; $F_l = F_{w,wB}$ = -42.7 kN

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Windward net force; $F_w = F_{w,wA} = 5.7$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 48.4$Kn

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	34	53.89
B	3.30	-0.50	1.29	0.16	33.99	5.47
C	3.30	-0.70	1.29	-0.06	13.86	-0.80
D	3.30	-0.70	1.29	-0.06	13.86	-0.80

Overall loading

Projected vertical plan area of wall;

$A_{vert_w_0} = b \times H = 33.99$ m^2

Projected vertical area of roof; $A_{vert_r_0} =$ 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} =$ 26.2 kN

Leeward net force; $F_l = F_{w,wB} = 5.5$ kN

Windward net force; $F_w = F_{w,wA} = 53.9$ kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 48.4$Kn

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	13.86	2.32
B	3.30	-0.23	1.29	-1.01	13.86	-14.03
C	3.30	-0.70	1.29	-1.48	33.99	-50.15
D	3.30	-0.70	1.29	-1.48	33.99	-50.15

Overall loading

Projected vertical plan area of wall;

$A_{vert_w_0} = d \times H = 13.86$ m^2

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Projected vertical area of roof; $A_{vert_r_0}$ = 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0}$ = 10.7 kN

Leeward net force; $F_l = F_{w,wB}$ = -14.0 kN

Windward net force; $F_w = F_{w,wA}$ =2.30 kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = 16.4Kn

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	13.86	21.97
B	3.30	-0.23	1.29	0.45	13.86	5.61
C	3.30	-0.70	1.29	-0.06	33.99	-1.97
D	3.30	-0.70	1.29	-0.06	33.99	-1.97

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = d \times H$ = 13.86 m^2

Projected vertical area of roof; $A_{vert_r_0}$ = 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0}$ = 10.7 kN

Leeward net force; $F_l = F_{w,wB}$ = 5.60 kN

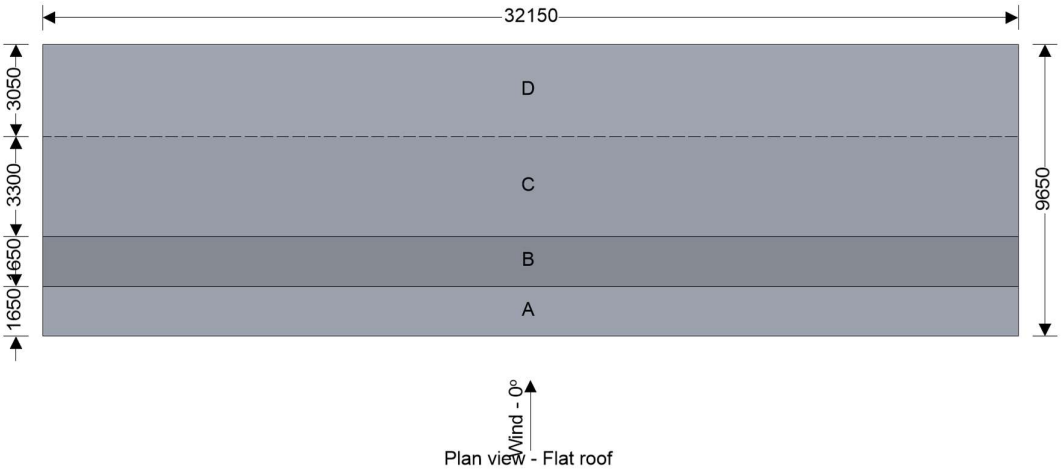
Windward net force; $F_w = F_{w,wA}$ =22.0 kN

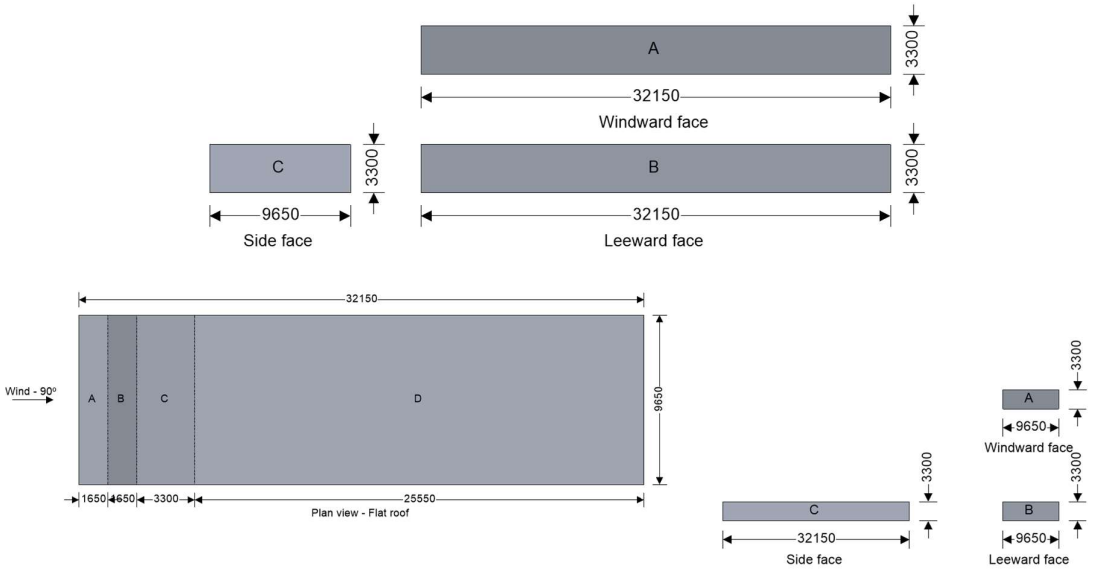
Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = 16.4Kn

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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4.5. Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

4.6. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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5. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

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Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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6. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

7. DEFLECTIONS

7.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support

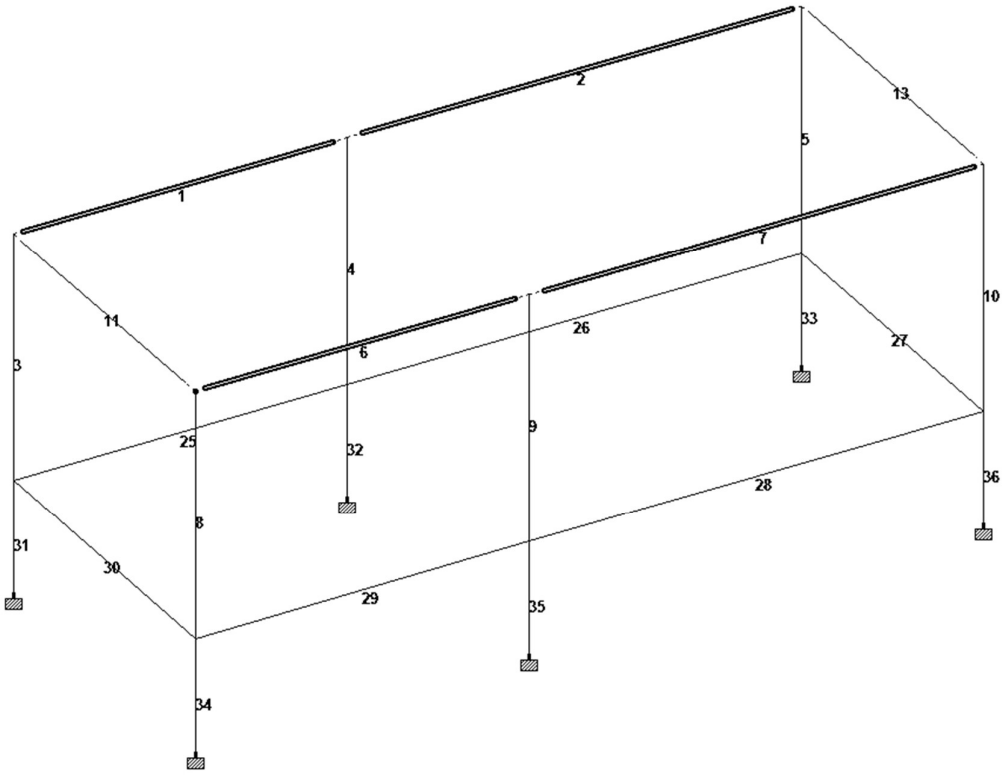
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8. CONCRETE ROOF BEAM DESIGN

8.1. Design of Ledger Beam(LB1)



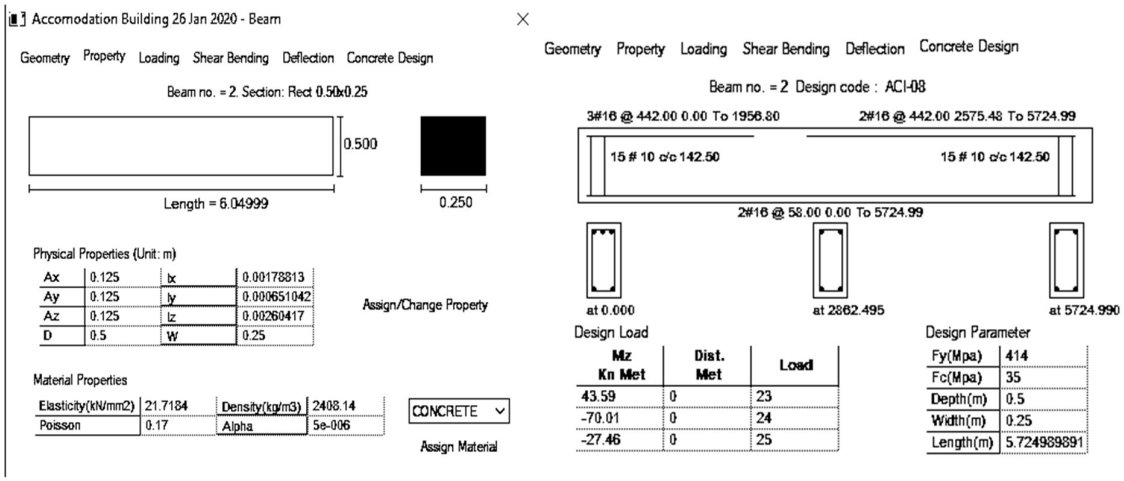
All | Summary | Envelope

	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
Max Fx	32	23 ULS: (1.4	24	**241.162**	0.202	11.326	0.036	6.506	-0.161
Min Fx	31	5 WL+X	1	**-19.552**	3.399	12.006	1.914	0.630	1.725
Max Fy	2	23 ULS: (1.4	5	14.503	**70.723**	-0.190	2.370	0.651	70.009
Min Fy	28	22 ULS: (1.4	8	4.795	**-57.652**	-0.033	0.645	-0.084	59.744
Max Fz	36	26 ULS: (1.2	9	167.616	21.397	**36.227**	-1.394	-15.965	19.402
Min Fz	33	23 ULS: (1.4	3	163.144	36.645	**-28.417**	0.005	23.164	30.512
Max Mx	7	27 ULS: (1.2	12	8.366	-49.070	2.828	**9.992**	6.996	26.320
Min Mx	6	26 ULS: (1.2	10	4.466	28.040	-4.757	**-11.237**	9.285	3.847
Max My	36	26 ULS: (1.2	28	171.867	21.397	36.227	-1.394	**38.376**	-12.693
Min My	36	23 ULS: (1.4	9	163.144	36.645	28.417	-0.005	**-23.164**	30.512
Max Mz	2	23 ULS: (1.4	5	14.503	70.723	-0.190	2.370	0.651	**70.009**
Min Mz	32	26 ULS: (1.2	24	212.048	26.220	-2.644	1.597	-7.835	**-39.164**

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Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **70 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

Ast.req = **485 mm^2**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm^2**

Maximum Moment required from Analysis = **43 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom considering Long term deflection check as shown below in section 9.1.1.

Ast.pro= **600mm^2 >>>>>>(Ast.min, Ast.req)**

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8.1.1. Long Term Deflection Check for LB1 Beam:

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa	
Reinforcement bar tensile stress, fy	=	414.00	MPa	
Modulus Elasticity of concrete,	=	27,806	Mpa	
Modulus Elasticity of steel, *Es*	=	200,000	Mpa	
Unit wt. of concrete, Wconc	=	24.00	kN/m^3	
Unit wt. of steel	=	77.00	kN/m^3	
Depth of beam, h	=	500	mm	
width of beam, b	=	250	mm	
Length of beam, L	=	6.00	m	
Bottom bar diam., db	=	16.00	mm	
area of 1 bar Ab	=	201.06	mm^2	
no. of bottom bars	=	3.00	pcs.	
Top bar diam., db	=	16.00	mm	
area of 1 bar Ab	=	201.06	mm^2	
no. of top bars	=	3.00	pcs.	
Stirrups diam. Dv	=	10.00	mm	
area of 1 bar Av	=	78.54	mm^2	
concrete cover, cc	=	40.00	mm	
$d_{eff.bot}$ (@ bottom), d	=	442.00	mm	
Mdead	=	52.00	kN-m	(FROM STAAD RESULTS)
M live	=	7.50	kN-m	

CHECK DEFLECTION: $M_{d+\ell}$ = 59.50 kN-m

$M_{sus} = M_d + 0.50M_\ell$ = 55.75 kN-m

modulus of rupture, modulus of elasticity, modular ratio

$fr = 0.62\lambda\sqrt{fc'}$ = 3.67 Mpa ←*ACI(318M-08) Sec. 9.5.2.3*

$n_s = \frac{Es}{Ec}$ = 7.19

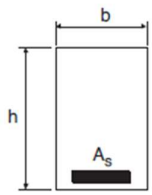
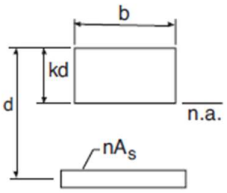
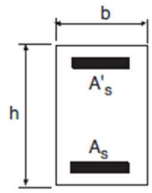
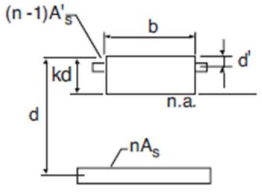
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 2.60E+09 \text{ mm}^4$$

$$B = \frac{b}{nAs} = 0.06 \text{ mm/mm}^2$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 107.71 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 5.89E+08 \text{ mm}^4$$

$$\frac{I_g}{I_{cr}} = 4.42$$

effective moment of inertia

$$M_{cr} = \frac{fr \cdot I_g}{y_t} = 38.21 \text{ kN-m}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead load

$$\frac{M_{cr}}{M_d} = 0.73$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 1.39E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.69$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 1.24E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.64$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 1.12E+09 \text{ mm}^4$$ ←*ACI(318M-08) Sec. 9.5.2.3*

initial or short-term deflection

$K = 1.00$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 5.051 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 6.075 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 7.148 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 2.097 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$.

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$(\Delta_i)_\ell \le \frac{\ell}{180}$ = 33.33 mm > 2.10 mm *ok!*

$(\Delta_i)_\ell \le \frac{\ell}{360}$ = 16.67 mm > 2.10 mm *ok!*

additional long-term deflection

ξ = 2.00

5 years or more 2.0
12 months..... 1.4
6 months..... 1.2
3 months..... 1.0

$\rho = \frac{As}{bd}$ = 0.0055

$\lambda_\Delta = \frac{\xi}{1+50\rho'}$ = 1.57 ←*ACI(318M-08) Sec. 9.5.2.5*

$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus}$ = 9.54 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell$ = 11.64 mm

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480}$ = 12.50 mm > 11.64 mm *ok!*

$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240}$ = 25.00 mm > 11.64 mm *ok!*

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8.1.2. Staad Output For LB1

```
        BEAM  NO.     2 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN -   6050. MM  FY -   414.   FC -   35.   MPA, SIZE -   250. X   500. MMS

LEVEL     HEIGHT       BAR INFO        FROM          TO          ANCHOR
            (MM)                         (MM)          (MM)      STA   END
 ------------------------------------------------------------------------

   1          58.      2 - 16MM          0.          6050.        YES   YES
   2         442.      3 - 16MM          0.          3046.        YES   NO
   3         442.      2 - 16MM       2752.          6050.        NO    YES


        B E A M  N O.     2 D E S I G N  R E S U L T S - SHEAR

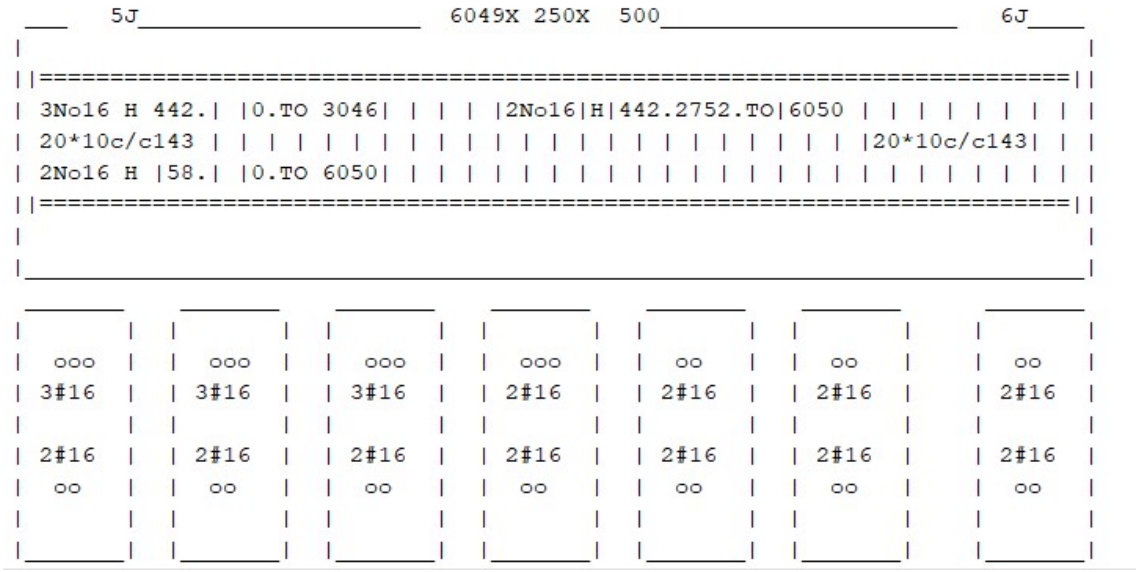
 AT START SUPPORT - Vu=    56.66 KNS   Vc=   107.61 KNS   Vs=     0.00 KNS
 Tu=     7.76 KN-MET   Tc=     3.9 KN-MET   Ts=    10.4 KN-MET   LOAD     27
 STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2585. MM
 ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    2.56 SQ.CM.
```


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```
AT END   SUPPORT - Vu=   48.03 KNS  Vc=  124.21 KNS  Vs=    0.00 KNS
Tu=    5.88 KN-MET  Tc=    3.9 KN-MET  Ts=    7.8 KN-MET  LOAD    23
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2585. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.94 SQ.CM.
```



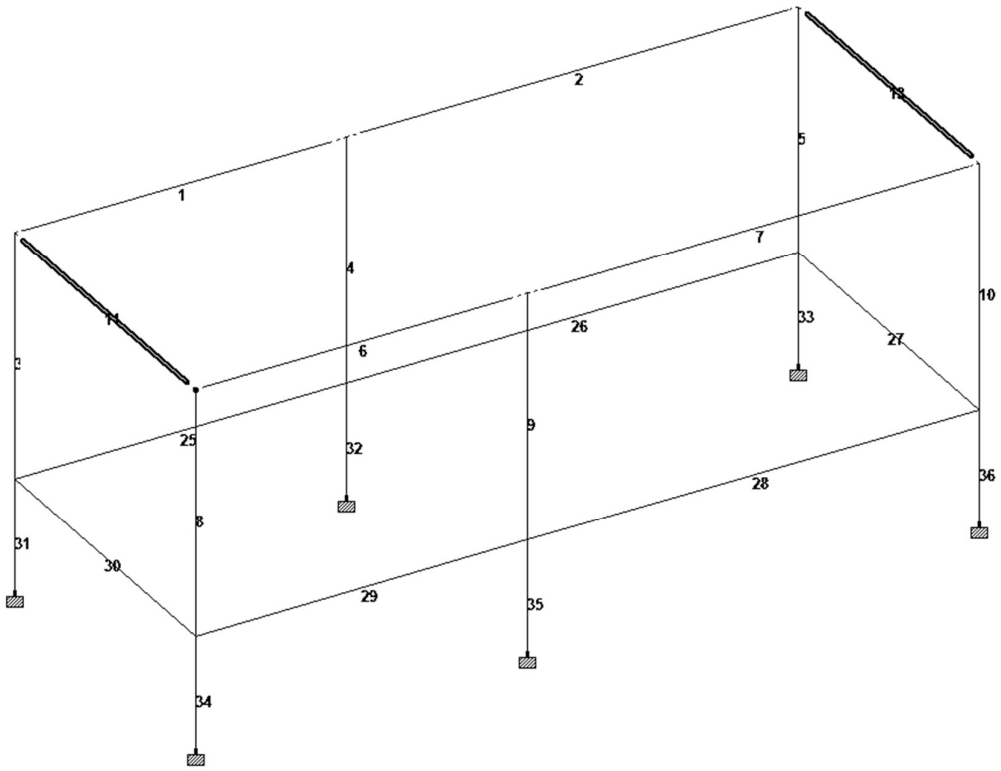
However Provide 3-16 dia-Full Straight Bars Top & Bottom considering Long term deflection check as shown below in section 9.1.1.

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8.2. Design of Beam-(B1)



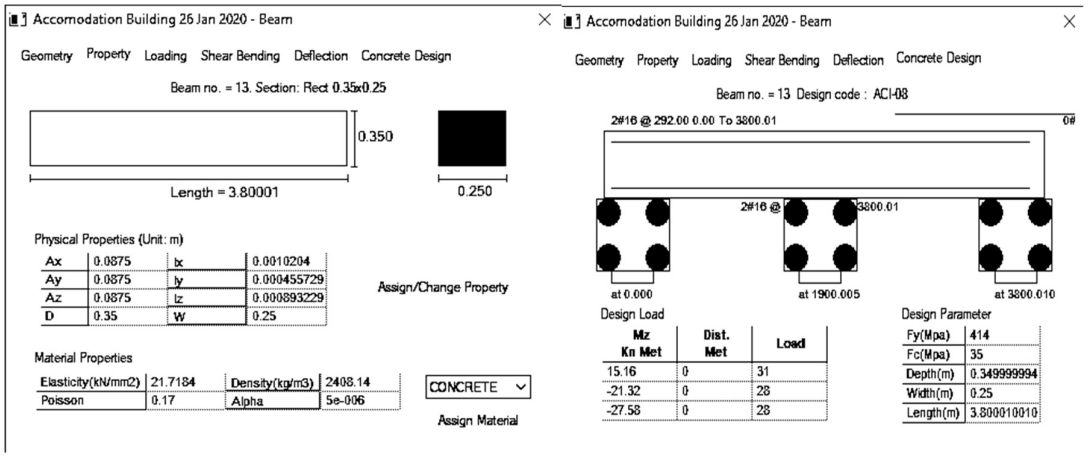
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Beam	Node	Env	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
11	4	+ve	2.392	6.075	1.676	0.000	0.608	4.896
			31 GENERAT	22 GENERAT	27 GENERAT	2 EQK LOAD	23 GENERAT	22 GENERAT
		-ve	-1.019	-5.089	-0.141	-3.353	-3.631	-14.067
			7 TEMP	5 WL+X	23 GENERAT	23 GENERAT	31 GENERAT	31 GENERAT
11	10	+ve	2.392	0.008	1.676	3.031	3.505	21.284
			31 GENERAT	7 TEMP	27 GENERAT	23 GENERAT	27 GENERAT	27 GENERAT
		-ve	-1.019	-13.350	-0.141	-0.135	-0.220	-0.207
			7 TEMP	27 GENERAT	23 GENERAT	7 TEMP	7 TEMP	7 TEMP
13	6	+ve	2.445	6.075	0.056	3.231	2.641	4.956
			31 GENERAT	22 GENERAT	23 GENERAT	23 GENERAT	31 GENERAT	22 GENERAT
		-ve	-0.996	-5.221	-1.218	-0.015	-0.349	-14.443
			7 TEMP	5 WL+X	27 GENERAT	5 WL+X	23 GENERAT	31 GENERAT
13	12	+ve	2.445	0.000	0.056	0.033	0.077	21.812
			31 GENERAT	2 EQK LOAD	23 GENERAT	7 TEMP	7 TEMP	27 GENERAT
		-ve	-0.996	-13.561	-1.218	-3.153	-2.546	-0.193
			7 TEMP	27 GENERAT	27 GENERAT	23 GENERAT	27 GENERAT	7 TEMP

MOMENT ENVELOPE FOR BEAMS B1



Design of Beam (B1)

Maximum Moment required from Analysis = **4.157 Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=350-40-10-12/2=294mm, Fc'=35 Mpa & ϕ=0.9

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Ast.req = **310 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **262mm²**

Maximum Moment required from Analysis = **13 Knm (Sagging)**

Hence Provide 3-12 dia-Full Straight Bars Top & Bottom

Ast.pro= **339mm² >>>>>>(Ast.min, Ast.req)**

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8.2.1. Staad Output For B1

```
        BEAM  NO .    12  DESIGN  RESULTS  - FLEXTURE  PER CODE  ACI 318 - 08

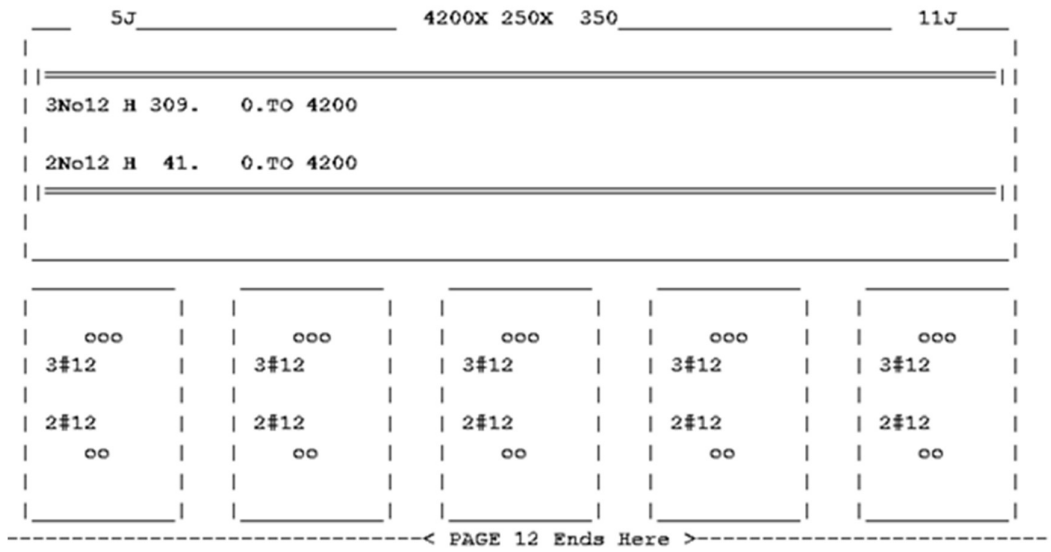
LEN -  4200. MM  FY -  414.  FC -  35.  MPA, SIZE -  250. X  350. MMS

LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
            (MM)                       (MM)          (MM)        STA  END
_____________________________________________________________________

  1          41.      2 - 12MM          0.          4200.        YES  YES
  2         309.      3 - 12MM          0.          4200.        YES  YES


        B E A M  N O.    12 D E S I G N  R E S U L T S - SHEAR

 AT START SUPPORT - Vu=     3.71 KNS  Vc=    75.82 KNS  Vs=     0.00 KNS
 Tu=     0.00 KN-MET   Tc=     2.3 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=     3.71 KNS  Vc=    75.82 KNS  Vs=     0.00 KNS
 Tu=     0.00 KN-MET   Tc=     2.3 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                   STIRRUPS ARE NOT REQUIRED.
```




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---------------------------------< PAGE 12 Ends Here >---------------------------
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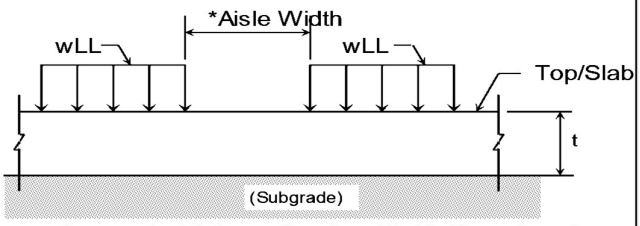
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9. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS
For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Low. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

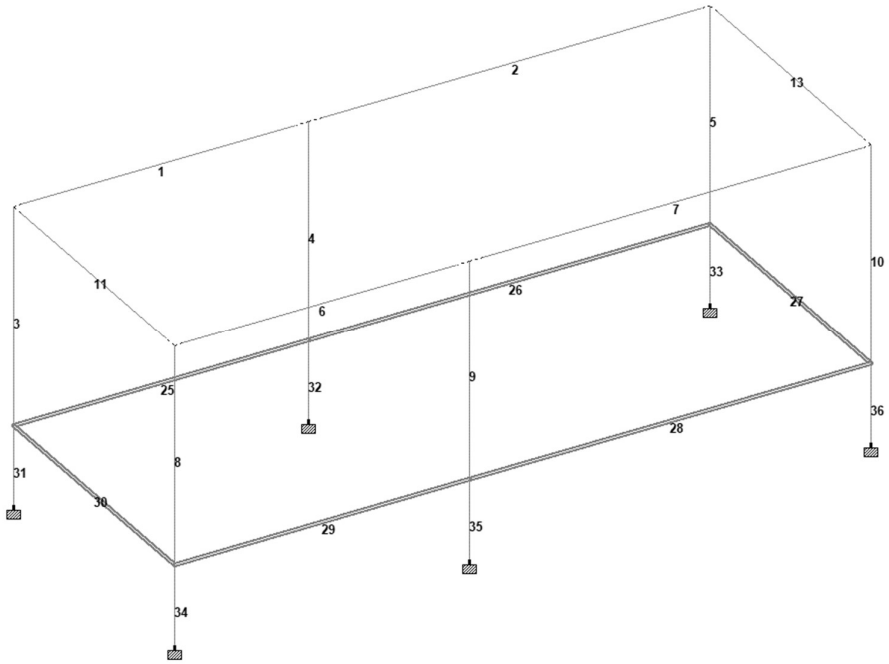
Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

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10. DESIGN OF BEAM-(GB1)



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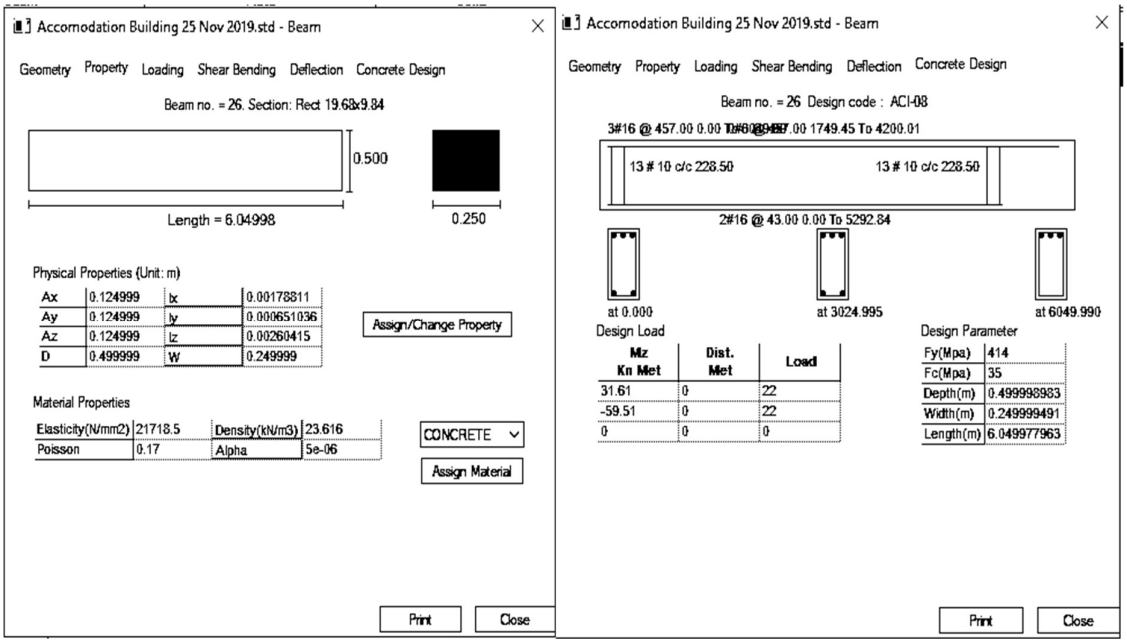
Beam	Node	Env	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
25	1	+ve	16.141	36.520	0.005	1.127	0.388	22.898
			23 GENERAT	22 GENERAT	23 GENERAT	27 GENERAT	31 GENERAT	23 GENERAT
		-ve	-0.654	-2.256	-0.225	-0.196	-0.011	-4.891
			4 LL	2 EQK LOAD	31 GENERAT	7 TEMP	23 GENERAT	2 EQK LOAD
25	2	+ve	16.141	1.521	0.005	1.127	0.013	38.615
			23 GENERAT	7 TEMP	23 GENERAT	27 GENERAT	23 GENERAT	27 GENERAT
		-ve	-0.654	-44.171	-0.225	-0.196	-0.614	-2.371
			4 LL	22 GENERAT	31 GENERAT	7 TEMP	31 GENERAT	7 TEMP
26	2	+ve	15.401	57.552	0.124	0.147	0.005	59.557
			23 GENERAT	22 GENERAT	31 GENERAT	7 TEMP	23 GENERAT	22 GENERAT
		-ve	-2.240	-1.316	-0.003	-1.192	-0.465	-3.989
			4 LL	2 EQK LOAD	23 GENERAT	27 GENERAT	31 GENERAT	2 EQK LOAD
26	3	+ve	15.401	0.000	0.124	0.147	0.289	43.020
			23 GENERAT	-	31 GENERAT	7 TEMP	31 GENERAT	22 GENERAT
		-ve	-2.240	-52.111	-0.003	-1.192	-0.007	0.000
			4 LL	22 GENERAT	23 GENERAT	27 GENERAT	23 GENERAT	-
27	3	+ve	24.066	38.079	0.000	0.000	0.569	23.415
			23 GENERAT	22 GENERAT	2 EQK LOAD	2 EQK LOAD	31 GENERAT	22 GENERAT
		-ve	0.000	-7.026	-0.271	-0.113	-0.002	-14.758
			-	5 WL+X	27 GENERAT	27 GENERAT	23 GENERAT	5 WL+X
27	9	+ve	24.066	0.000	0.000	0.000	0.000	43.634
			23 GENERAT	2 EQK LOAD	2 EQK LOAD	2 EQK LOAD	2 EQK LOAD	27 GENERAT
		-ve	0.000	-43.881	-0.271	-0.113	-0.570	-0.039
			-	27 GENERAT	27 GENERAT	27 GENERAT	27 GENERAT	4 LL
28	9	+ve	15.401	52.111	0.127	0.532	0.000	43.020
			23 GENERAT	22 GENERAT	27 GENERAT	22 GENERAT	2 EQK LOAD	22 GENERAT
		-ve	-2.240	-0.018	-0.000	-0.462	-0.296	-0.055
			4 LL	1 EQK LOAD	2 EQK LOAD	5 WL+X	27 GENERAT	1 EQK LOAD
28	8	+ve	15.401	1.316	0.127	0.532	0.473	59.597
			23 GENERAT	2 EQK LOAD	27 GENERAT	22 GENERAT	27 GENERAT	22 GENERAT
		-ve	-2.240	-57.592	-0.000	-0.462	-0.000	-3.989
			4 LL	22 GENERAT	2 EQK LOAD	5 WL+X	2 EQK LOAD	2 EQK LOAD
29	8	+ve	16.141	44.171	0.000	0.336	0.625	37.732
			23 GENERAT	22 GENERAT	-	5 WL+X	27 GENERAT	27 GENERAT
		-ve	-0.654	-1.521	-0.230	-0.696	0.000	-2.371
			4 LL	7 TEMP	27 GENERAT	22 GENERAT	-	7 TEMP
29	7	+ve	16.141	2.256	0.000	0.336	0.004	22.898
			23 GENERAT	2 EQK LOAD	-	5 WL+X	1 EQK LOAD	23 GENERAT
		-ve	-0.654	-36.520	-0.230	-0.696	-0.398	-4.891
			4 LL	22 GENERAT	27 GENERAT	22 GENERAT	27 GENERAT	2 EQK LOAD
30	7	+ve	23.463	42.226	0.430	0.000	0.000	40.225
			23 GENERAT	27 GENERAT	27 GENERAT	2 EQK LOAD	-	27 GENERAT
		-ve	0.000	-0.000	-0.000	-0.096	-0.904	-0.033
			-	4 LL	23 GENERAT	27 GENERAT	27 GENERAT	4 LL
30	1	+ve	23.463	5.992	0.430	0.000	0.901	23.484
			23 GENERAT	5 WL+X	27 GENERAT	2 EQK LOAD	31 GENERAT	22 GENERAT
		-ve	0.000	-38.079	-0.000	-0.096	-0.002	-12.585

MOMENT ENVELOPE FOR BEAMS GB1

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Maximum Moment required from Analysis = **59 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-50-10-16/2=432mm, Fc'=35 Mpa & ϕ=0.9

Ast.req = **375 mm²**

Ast.min= $$A_{s,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **31 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

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STAAD OUTPUT FOR GB1

```
        BEAM  NO.    26  DESIGN  RESULTS   -  FLEXURE   PER  CODE ACI  318-08

 LEN -   6050. MM   FY -   414.   FC -   35.   MPA, SIZE -   250. X   500. MMS

 LEVEL     HEIGHT      BAR INFO        FROM          TO          ANCHOR
             (MM)                        (MM)          (MM)        STA   END
 ____________________________________________________________________________


   1          43.      2 - 16MM          0.          5293.         YES   NO
   2         457.      3 - 16MM          0.          6050.         YES   YES


        B E A M  N O.    26 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=    49.34 KNS   Vc=  109.44 KNS   Vs=     0.00 KNS
  Tu=     0.53 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD     22
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 228. MM  C/C FOR 2570. MM

  AT END    SUPPORT - Vu=    43.86 KNS   Vc=  110.84 KNS   Vs=     0.00 KNS
  Tu=     0.53 KN-MET   Tc=     3.9 KN-MET   Ts=     0.0 KN-MET   LOAD     22
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 228. MM  C/C FOR 2570. MM

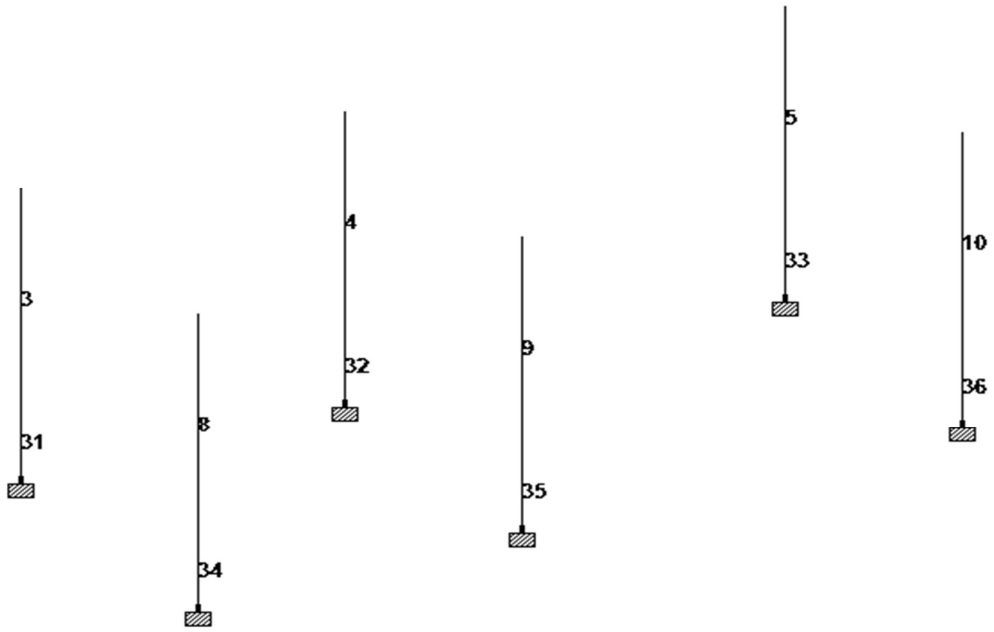

 ___    2J_____________________   6049X 250X  500_____________________    3J____
|                                                                                |
||================================================================================||
| 3No16 H 457.|  0.TO 6050|  |  |  |  |  |  |  |  |  |  |  |  |  |            |
| 13*10c/c229 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13*10c/c229    |
| 2No16 H  43.|  0.TO 5293|  |  |  |  |  |  |  |  |  |  |  |  |  |            |
||================================================================             |
|                                                                                |
|________________________________________________________________________________|
 _______          _______          _______          _______          _______
|       |        |       |        |       |        |       |        |       |
|  ooo  |        |  ooo  |        |  ooo  |        |  ooo  |        |  ooo  |
| 3#16  |        | 3#16  |        | 3#16  |        | 3#16  |        | 3#16  |
|       |        |       |        |       |        |       |        |       |
| 2#16  |        | 2#16  |        | 2#16  |        | 2#16  |        |       |
|  oo   |        |  oo   |        |  oo   |        |  oo   |        |       |
|       |        |       |        |       |        |       |        |       |
|_______|        |_______|        |_______|        |_______|        |_______|
```


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11. CONCRETE COLUMN DESIGN



Column numbers

Beam End Force Summary

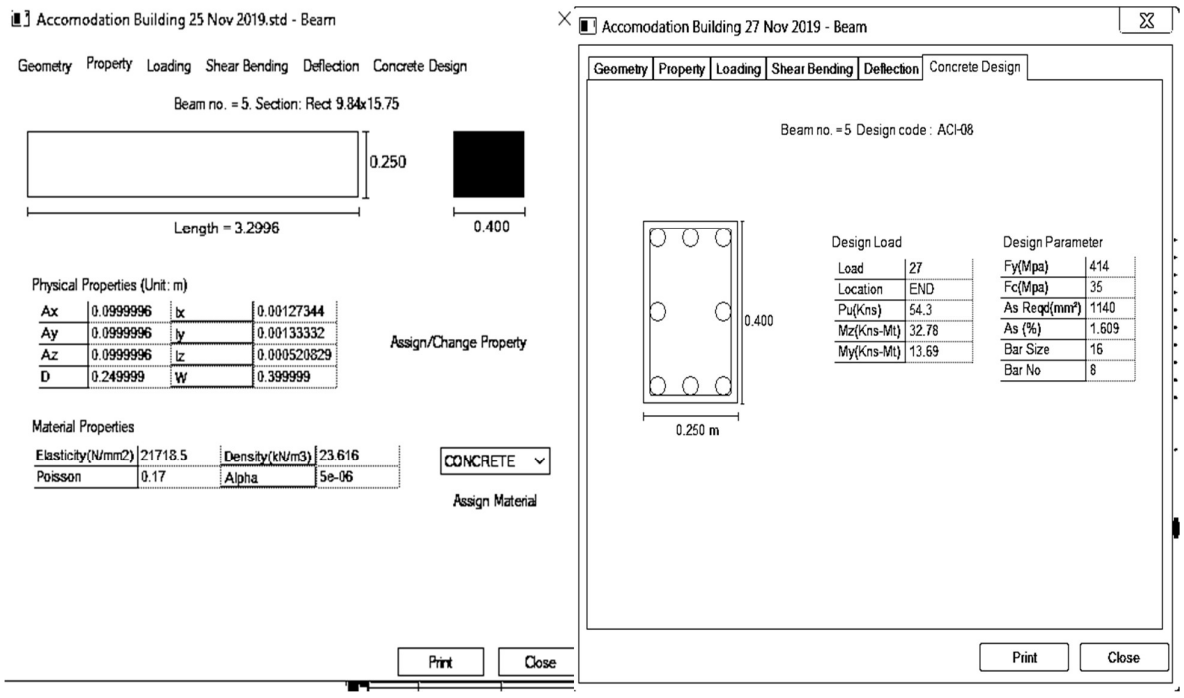
The signs of the forces at end B of each beam have been reversed. For example: this means that the Min Fx entry gives the largest tension value for an beam.

	Beam	Node	L/C	Axial Fx (kN)	Shear Fy (kN)	Shear Fz (kN)	Torsion Mx (kNm)	Bending My (kNm)	Bending Mz (kNm)
Max Fx	32	24	23:ULS: (1.4D+	**241.162**	0.202	11.326	0.036	6.506	-0.161
Min Fx	31	1	5:WL+X	**-19.552**	3.399	12.006	1.914	0.630	1.725
Max Fy	2	5	23:ULS: (1.4D+	14.503	**70.723**	-0.190	2.370	0.651	70.009
Min Fy	28	8	22:ULS: (1.4 D	4.795	**-57.652**	-0.033	0.645	-0.084	59.744
Max Fz	36	9	26:ULS: (1.2D+	167.616	21.397	**36.227**	-1.394	-15.965	19.402
Min Fz	33	3	23:ULS: (1.4D+	163.144	36.645	**-28.417**	0.005	23.164	30.512
Max Mx	7	12	27:ULS: (1.2D+	8.366	-49.070	2.828	**9.992**	6.996	26.320
Min Mx	6	10	26:ULS: (1.2D+	4.466	28.040	-4.757	**-11.237**	9.285	3.847
Max My	36	28	26:ULS: (1.2D+	171.867	21.397	36.227	-1.394	**38.376**	-12.693
Min My	36	9	23:ULS: (1.4D+	163.144	36.645	28.417	-0.005	**-23.164**	30.512
Max Mz	2	5	23:ULS: (1.4D+	14.503	70.723	-0.190	2.370	0.651	**70.009**
Min Mz	32	24	26:ULS: (1.2D+	212.048	26.220	-2.644	1.597	-7.835	**-39.164**

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Longitudinal Reinforcement, Ast.required from Analysis = **1140 mm2**

Provide 6 No's-16 Dia Vertical Bars, $A_{st.pro}$= **1200mm² >>>>1140mm2**

11.1. Staad Output For Column C1

```
================================================================

      COLUMN   NO.      5   DESIGN PER ACI 318-08 - AXIAL + BENDING


 FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 400.0 X 250.0 MMS, TIED
         AREA OF STEEL REQUIRED =   1140.0   SQ. MM

 BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
 ---------------------------------------------------------------

--------------------------------< PAGE 21 Ends Here >----------------------------
  STAAD SPACE                                                  -- PAGE NO.    22

   8 - 16 MM                 1.609        27       END      0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER    10 SPACING 250.00 MM


================================================================
```


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12. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties			
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r"	24		kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35		N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420		N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20		mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9		
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75		
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es'	2E+05		N/mm^2
Loading			Modulus of Elasticity of concrete	27806		N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19		
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**			
Wall load	0	kN/m^2	Top layer Nos.	6	Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0	Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"		468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"		0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db' = 10 mm			
Calculations						
Shear force Vu = U*le	**8.096**	**kN**				
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**				

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm					
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm	
Moment due to live load "Mll"	0.32	kNm	
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm	
Deflection due to dead + live loads			
R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411		
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4	
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm	

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b **$I_e/360$ 2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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13. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES:

BUILDING PLANS	MD-216-6000-EG-AR-GAD-3087
OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-GAD-1511
LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-GAD-1512
GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0.122 0 0; 2 4.57201 0 0; 3 10.622 0 0; 4 0.122 3.29961 0;
5 4.57201 3.29961 0; 6 10.622 3.29961 0; 7 0.122 0 4.20001;
8 4.57201 0 4.20001; 9 10.622 0 4.20001; 10 0.122 3.29961 4.20001;
11 4.57201 3.29961 4.20001; 12 10.622 3.29961 4.20001; 23 0.122 -1.5 0;
24 4.57201 -1.5 0; 25 10.622 -1.5 0; 26 0.122 -1.5 4.20001;
27 4.57201 -1.5 4.20001; 28 10.622 -1.5 4.20001;
MEMBER INCIDENCES
1 4 5; 2 5 6; 3 1 4; 4 2 5; 5 3 6; 6 10 11; 7 11 12; 8 7 10; 9 8 11; 10 9 12;
11 4 10; 13 6 12; 25 1 2; 26 2 3; 27 3 9; 28 9 8; 29 8 7; 30 7 1; 31 1 23;
32 2 24; 33 3 25; 34 7 26; 35 8 27; 36 9 28;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 TO 5 8 TO 10 31 TO 36 PRIS YD 0.25 ZD 0.4

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1 2 6 7 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

25 TO 30 PRIS YD 0.5 ZD 0.25

MEMBER PROPERTY AMERICAN

11 13 PRIS YD 0.35 ZD 0.25

CONSTANTS

BETA 90 MEMB 4 9 32 35

MATERIAL CONCRETE ALL

SUPPORTS

23 TO 28 FIXED

MEMBER CRACKED

3 TO 5 8 TO 10 31 TO 36 REDUCTION RIY 0.700000 RIZ 0.700000

1 2 6 7 11 13 25 TO 30 REDUCTION RIY 0.350000 RIZ 0.350000

MEMBER OFFSET

2 7 START 0.2 0 0

2 7 END -0.125 0 0

1 6 START 0.125 0 0

1 6 END -0.2 0 0

11 13 START 0 0 0.2

11 13 END 0 0 -0.2

DEFINE IBC 2003

SDS 0.1 SD1 1.36 S1 0.02 IE 1 RX 3 RZ 3 SCLASS 2

SELFWEIGHT 1

MEMBER WEIGHT

25 TO 30 UNI 10

MEMBER WEIGHT

1 2 6 7 UNI 8.9

DEFINE WIND LOAD

TYPE 1 WIND X WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

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2 4.000 M 10.000 M 5.000 M 2.000 0.010 0 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -

0 0 0 0.900 0.800 -0.180

!> END GENERATED DATA BLOCK

INT 1.16821 1.16821 0.46267 HEIG 0 4.572 4.57201

EXP 0.8 JOINT 1 TO 12 23 TO 28

TYPE 2 WIND X LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 10.000 FT 6.000 FT 2.000 0.010 1 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -

0 0 0 0.905 -0.300 0.550

!> END GENERATED DATA BLOCK

INT -1.06679 -1.06679 -0.456857 HEIG 0 4 4.00001

EXP 0.8 JOINT 1 TO 12 23 TO 28

TYPE 3 WIND Z WINDWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 6.000 M 10.000 M 2.000 0.010 0 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -

0 0 0 0.892 0.800 -0.550

!> END GENERATED DATA BLOCK

INT 1.64086 1.64086 0.650654 HEIG 0 4.572 4.57201

EXP 0.8 JOINT 1 TO 12 23 TO 28

TYPE 4 WIND Z LEEWARD

<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!

ASCE-7-2010:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -

1 4.000 M 6.000 M 10.000 M 2.000 0.010 1 -

0 0 0 0 1.030 1.000 1.000 0.850 0 -

0 0 0 0.892 -0.500 0.550

!> END GENERATED DATA BLOCK

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INT -1.29332 -1.29332 -0.513946 HEIG 0 4 4.00001

EXP 0.8 JOINT 1 TO 12 23 TO 28

LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z

IBC LOAD Z 1

PERFORM ANALYSIS

CHANGE

LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X

IBC LOAD X 1

PERFORM ANALYSIS

CHANGE

LOAD 3 LOADTYPE Dead TITLE DL

SELFWEIGHT Y -1

MEMBER LOAD

25 TO 30 UNI GY -10

MEMBER LOAD

1 2 6 7 UNI GY -12

11 UMOM GZ 1

13 UMOM GZ -1

1 2 UMOM GX -1

6 7 UMOM GX 1

LOAD 4 LOADTYPE Live TITLE LL

MEMBER LOAD

1 2 6 7 UNI GY -2.1

11 UMOM GZ 0.2

13 UMOM GZ -0.2

1 2 UMOM GX -0.2

6 7 UMOM GX 0.2

LOAD 5 LOADTYPE Wind TITLE WL+X

WIND LOAD X 1 TYPE 1 XR 0 0.2 YR 0 4

WIND LOAD -X 1 TYPE 2 XR 10.622 11 YR 0 4

WIND LOAD Z 1 TYPE 3 YR 0 4 ZR 0 0.1

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WIND LOAD -Z 1 TYPE 4 YR 0 4

LOAD 6 LOADTYPE Wind TITLE WL+Z

WIND LOAD Z 1 TYPE 3 XR -0.5 11.5 YR 0 4 ZR 0 0.1

WIND LOAD -Z 1 TYPE 4 YR 0 4 ZR 4.2 4.3

WIND LOAD X 1 TYPE 1 YR 0 4 ZR -0.6 4.8

WIND LOAD -X 1 TYPE 2 YR 0 4 ZR -0.6 4.8

LOAD 7 LOADTYPE Temperature TITLE TEMP

TEMPERATURE LOAD

1 TO 11 13 25 TO 36 TEMP 25

LOAD COMB 8 SLS: (DL)

3 1.0

LOAD COMB 9 SLS: (DL+T)

3 1.0 7 1.0

LOAD COMB 10 SLS: (D+LR)

3 1.0 4 1.0

LOAD COMB 11 SLS: (D+0.75LR)

3 1.0 4 0.75

LOAD COMB 12 SLS: (D+0.7EZ)

3 1.0 1 0.7

LOAD COMB 13 SLS: (D+0.7EX)

3 1.0 2 0.7

LOAD COMB 14 SLS: (D+0.75WX+0.75LR)

3 1.0 5 0.75 4 0.75

LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)

3 1.0 6 0.75 4 0.75

LOAD COMB 16 SLS: (0.6D+WX)

3 0.6 5 1.0

LOAD COMB 17 SLS: (0.6D+WZ)

3 0.6 6 1.0

LOAD COMB 18 SLS: (0.6D+0.7EX)

3 0.6 2 0.7

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LOAD COMB 19 SLS: (0.6D+0.7EZ)

3 0.6 1 0.7

LOAD COMB 20 SLS: (D+WX)

3 1.0 5 1.0

LOAD COMB 21 SLS: (D+WZ)

3 1.0 6 1.0

LOAD COMB 22 ULS: (1.4 DL)

3 1.4

LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)

3 1.4 4 0.5 7 1.4

LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)

3 1.2 5 0.8 4 1.6

LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 6 1.6 4 0.5

LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)

3 1.2 4 0.5 5 1.6

LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)

3 1.2 4 0.5 6 1.6

LOAD COMB 29 ULS: (1.2D+EX)

3 1.2 2 1.0

LOAD COMB 30 ULS: (1.2D+EZ)

3 1.2 1 1.0

LOAD COMB 31 ULS: (0.90D+1.6WX)

3 0.9 5 1.6

LOAD COMB 32 ULS: (0.90D+1.6WZ)

3 0.9 6 1.6

LOAD COMB 33 ULS: (0.90D+EZ)

3 0.9 1 1.0

LOAD COMB 34 ULS: (0.90D+EX)

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```
3 0.9 2 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 ALL
EFACE 0.44 MEMB 1 2 6 7
SFACE 0.44 MEMB 1 2 6 7
EFACE 0.29 MEMB 11 13
SFACE 0.29 MEMB 11 13
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MAXMAIN 16 ALL
MINMAIN 16 ALL
MINSEC 10 ALL
DESIGN BEAM 1 2 6 7 11 13 25 TO 30
DESIGN COLUMN 3 TO 5 8 TO 10 31 TO 36
END CONCRETE DESIGN
FINISH
```


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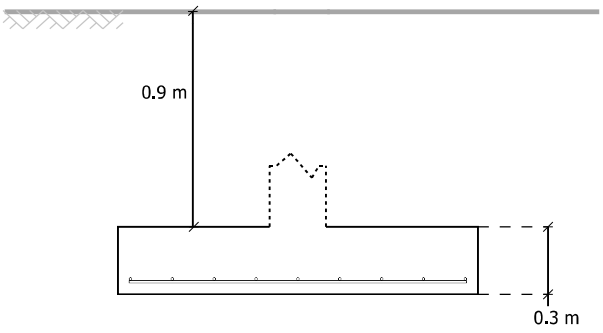
ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

Isolated Footing Design(ACI 318-08)

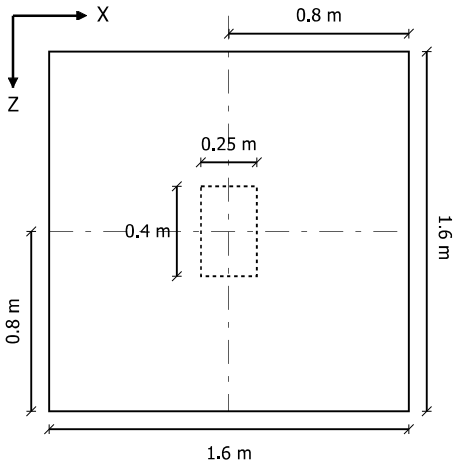
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
25	1	1.600m	1.600m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
25	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1 ALONG GRID 1 & C



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1600.000mm
Footing Width - Z (Fw) : 1600.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.400m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 50.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.600m

Initial Width (W_o) = 1.600m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
9	118.175	26.383	-20.232	-13.873	-17.561
10	122.688	13.965	-11.467	-5.194	-6.657
11	121.359	14.111	-11.420	-5.174	-6.721
12	113.344	14.582	-8.195	-0.425	-6.952
13	118.933	16.373	-11.281	-5.114	-8.978
14	123.138	16.739	-10.005	-4.595	-9.524
15	106.177	13.859	1.492	13.558	-6.427
16	72.794	12.232	-4.881	-2.297	-7.886
17	50.180	8.392	10.447	21.908	-3.757
18	71.985	10.555	-6.769	-3.068	-6.211
19	66.396	8.763	-3.683	1.620	-4.185
20	119.742	18.051	-9.394	-4.342	-10.653
21	97.128	14.211	5.935	19.862	-6.523

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
22	164.319	20.365	-15.794	-7.160	-9.682
23	168.103	36.645	-28.417	-19.462	-24.456
24	168.103	36.645	-28.417	-19.462	-24.456
25	151.250	19.329	-12.324	-5.647	-10.874
26	111.115	16.628	13.916	33.785	-7.541

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
27	147.298	22.772	-10.610	-4.942	-14.148
28	111.115	16.628	13.916	33.785	-7.541
29	143.077	20.065	-13.537	-6.137	-11.244
30	135.092	17.506	-9.129	0.561	-8.350
31	109.428	18.699	-7.133	-3.368	-12.203
32	73.246	12.554	17.393	35.359	-5.596
33	99.881	13.142	-5.745	2.095	-6.275
34	107.866	15.702	-10.153	-4.603	-9.170

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.560 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 0.933 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.600 m Governing Load Case : # 9

Width (W_2) = 1.600 m Governing Load Case : # 9

Depth (D_2) = 0.300 m Governing Load Case : # 23

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.560 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 18.431 kN

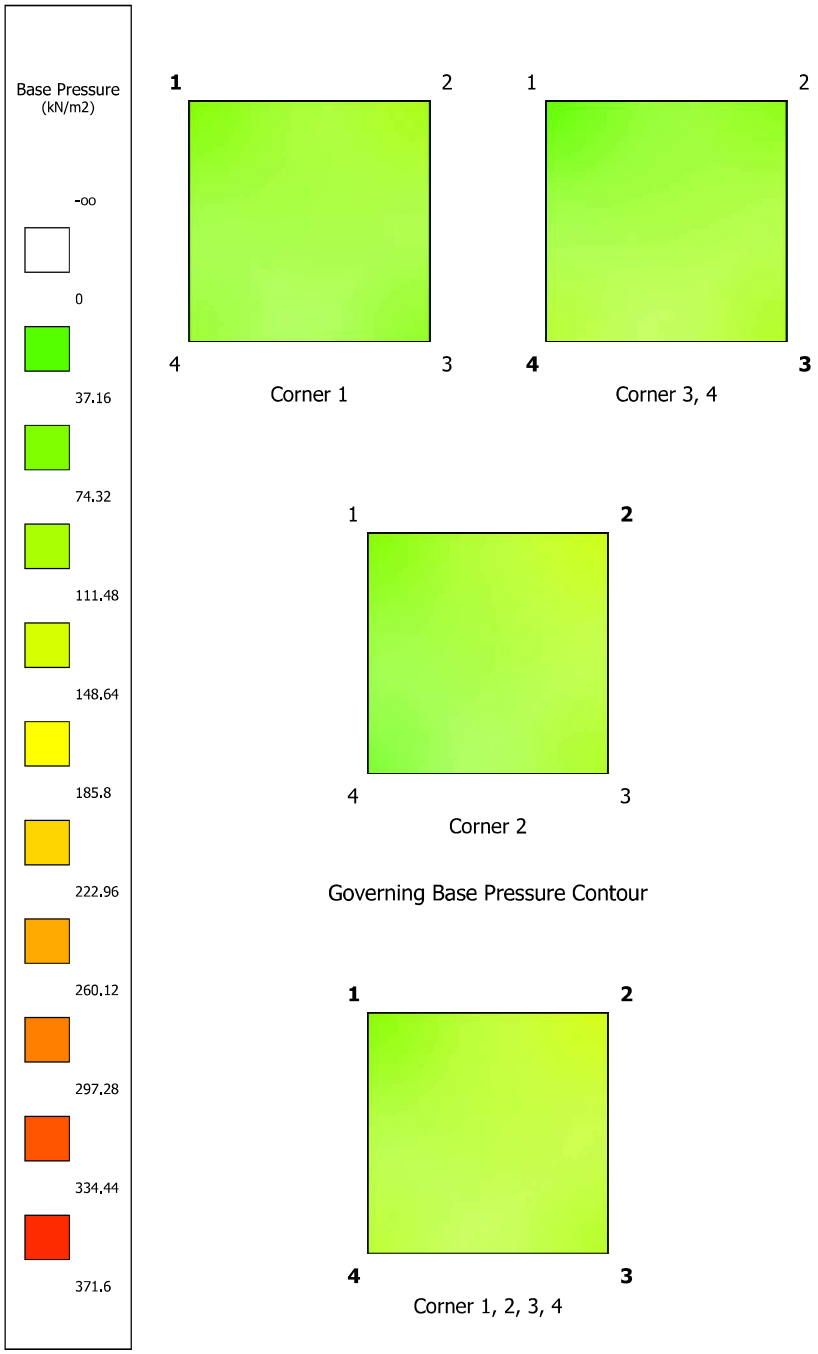
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 39.851 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

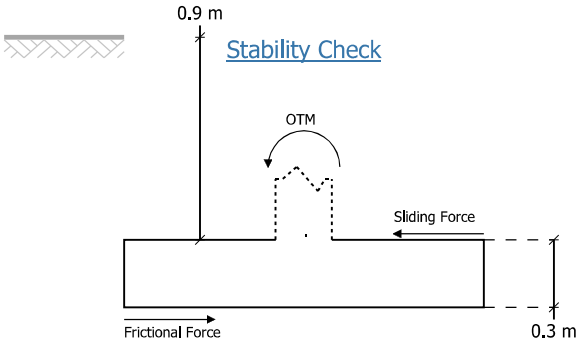


Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
10	**67.4505**	99.2269	73.9323	42.1559	0.000
9	60.8222	**135.4600**	77.0346	2.3968	0.000
21	13.2031	44.8042	**108.2111**	76.6100	0.000
21	13.2031	44.8042	108.2111	**76.6100**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
10	**67.4505**	99.2269	73.9323	42.1559
9	60.8222	**135.4600**	77.0346	2.3968
21	13.2031	44.8042	**108.2111**	76.6100
21	13.2031	44.8042	108.2111	**76.6100**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
9	3.879	5.059	4.353	1.500	7.092	5.576	1.500
10	7.516	9.154	8.215	1.500	16.798	13.430	1.500
11	7.384	9.123	8.117	1.500	16.741	13.200	1.500
12	6.827	12.146	8.416	1.500	47.696	12.201	1.500
13	6.278	9.111	7.311	1.500	16.712	10.271	1.500
14	6.286	10.517	7.631	1.500	19.140	10.039	1.500
15	6.883	63.937	9.678	1.500	9.412	12.514	1.500
16	6.215	15.574	8.163	1.500	27.950	9.151	1.500
17	7.496	6.021	6.639	1.500	3.475	13.971	1.500
18	7.158	11.162	8.522	1.500	20.488	11.208	1.500
19	8.252	19.635	10.759	1.500	194.041	14.768	1.500
20	5.720	10.992	7.176	1.500	19.926	8.919	1.500
21	6.343	15.188	8.277	1.500	5.757	11.609	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 9
Governing Disturbing Force : 26.383kN
Governing Restoring Force : 102.345kN
Minimum Sliding Ratio for the Critical Load Case : 3.879
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 25.042kNm
Governing Resisting Moment : 87.027kNm
Minimum Overturning Ratio for the Critical Load Case : 3.475

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 9

Governing Disturbing Force : -20.232kN

Governing Restoring Force : 102.345kN

Minimum Sliding Ratio for the Critical Load Case : 5.059

Critical Load Case for Overturning about Z-Direction : 9

Governing Overturning Moment : -25.476kNm

Governing Resisting Moment : 142.054kNm

Minimum Overturning Ratio for the Critical Load Case : 5.576

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 9

Governing Disturbing Force : 33.247kN

Governing Restoring Force : 144.737kN

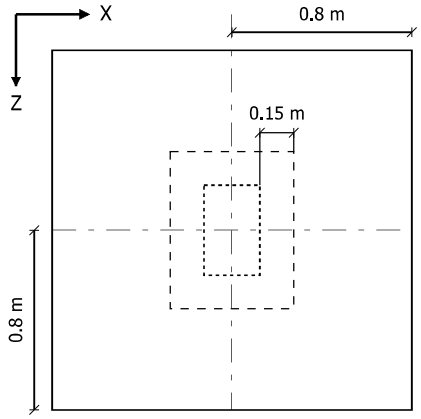
Minimum Sliding Ratio for the Critical Load Case : 4.353

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.600

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 200.776kN, Load Case # 23

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1227.632kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1715.472kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1091.228kN

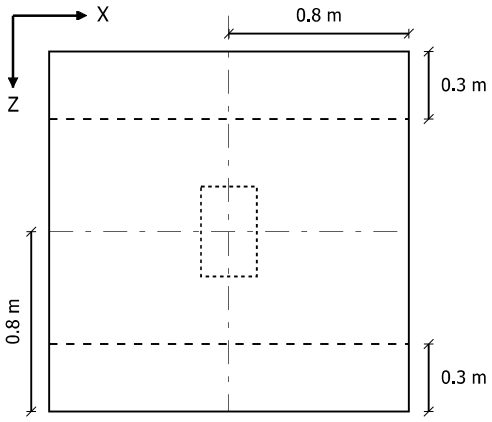
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 818.421kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

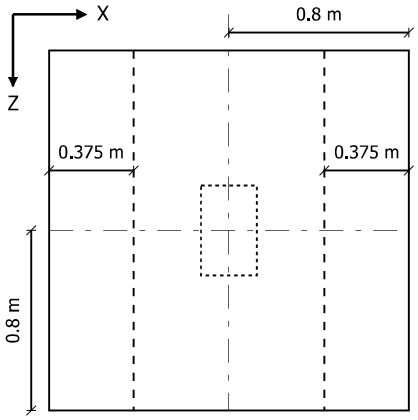
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{ux} is # 23 $V_{ux} = V_{ux}|_{x=D_x}$ = 68.527 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

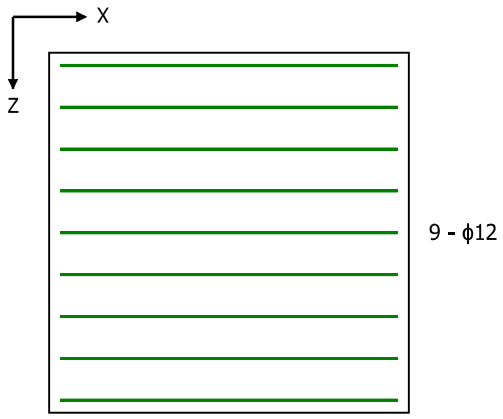
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 87.142 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00160
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.925 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	45.852 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	50.946 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00129
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00105
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	769.450 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 50.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.625 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.244 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00261

From ACI Cl.7.6.1, minimum req'd clear distance between bars

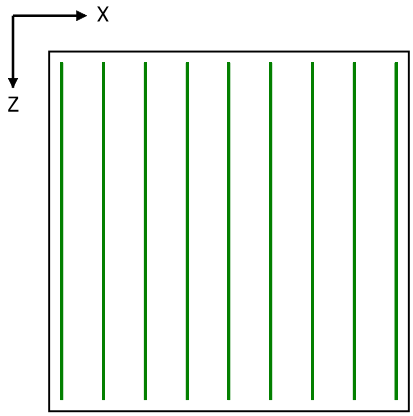
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.323

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 23

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.232 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00157

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.600 m

Ultimate moment, $M_u|_{x=D_x}$ = 34.321 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 38.134 kNm

(Based on effective depth) Required $\rho = \frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00106

(Based on gross depth) ρ x d_{eff} / Depth = 0.00082

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 755.524 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 50.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.232 m

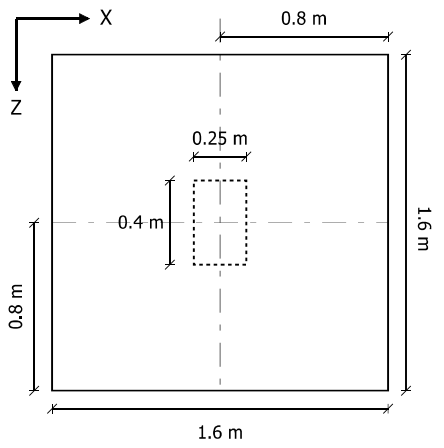
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00274

From ACI Cl.7.6.1, minimum req'd clear distance between bars

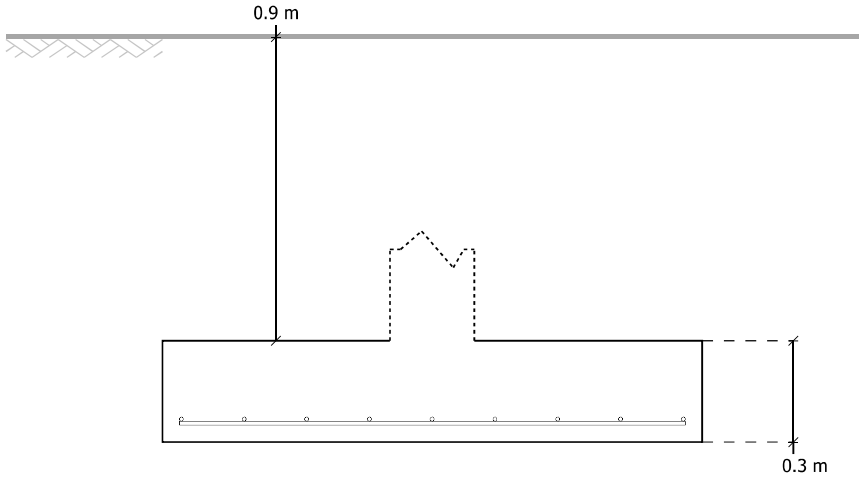
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 50.000mm

Provided Steel Area / Required Steel Area = 1.347

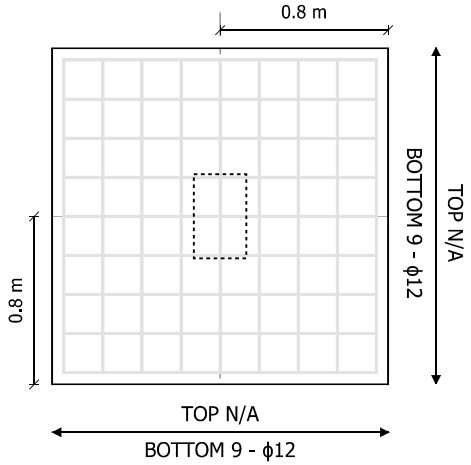
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.6 m	1.6 m	0.3 m	N/A	N/A	9 - ϕ12	9 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

ATTACHMENT 3 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT** **PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.** **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-001060, 17 January 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 17 January 2020	Comment by	EPC Response DATE: 02 March,2020	Status	Remarks
			General Comments (cc)				
MD-216-6000-EG-CV-CAL-1477-A01	6000	1	Please use the Structural Design Criteria belonging to this project Consider as primary codes all SBC's	Mr.Klaus Grof	Noted. Doc:MD-216-0000-EG-ST-CRT-0001 has been considered in the Analysis & Design. Saudi Building Code references are shown & highlighted in the Design Report & also some of the equivalent ACI codes are used where ever applicable.		
MD-216-6000-EG-CV-CAL-1477-A01	6000	2	Please establish in this document that all these wind calculations agree with SBC 301	Mr.Klaus Grof	Noted. Analythical Verification has been perfomed to check the Wind Loads generated in the Analysis with reference to SBC-301 & has been included in this revised submittal(A02)		
MD-216-6000-EG-CV-CAL-1477-A01	6000	3	Please consider SBC 301	Mr.Klaus Grof	Noted & References of SBC-301 have been provided where ever considered.		